



Python and Instant GUIs

Python has quite a few GUI tool-kits for creating large and powerful applications. But what if you want to quickly hack together a small script, without having to learn the API for a full-fledged GUI tool-kit? Read on...

There is a plethora of GUI tool-kits like GTK+, Qt, wxWidgets, FLTK, etc, which are available for different programming and scripting languages. Developing GUIs with scripting languages saves you the tedious compile-link-run cycles of languages like C, C++, etc, besides letting you run the same code on multiple platforms. Python, one of the most popular scripting languages, is great for GUI programming because of its elegance and since it has bindings for almost all popular GUI tool-kits. But most of these require a high level of initial learning, an unnecessary hurdle for when your needs are simple.

Most tool-kits are based on an event-driven model; applications built on these don't follow any predefined sequence of actions, but depend on the user's interaction with widgets in the application. In contrast, there's a model where you determine the presentation sequence for all widgets, or a grouping of widgets, for some general or frequently-required operation. The user interacts with these one at a time, and the application flow is strictly serial, from start to end. This is very useful for scripting short task-flows that proceed through the same steps each time, but with varying user inputs.

A third consideration is that the heavy dependencies

of some tool-kits may make the distribution of simple applications problematic or 'bloated'. Sometimes you only need a minimalist approach—to put a few widgets together for a quick and easy GUI, without any hassles. Here are a couple of possible solutions.

EasyGUI is a wrapper over Python's official GUI tool-kit, Tkinter; PyZenity wraps the GTK+ dialogue box's utility, Zenity. These provide the most common widgets used in serial-flow GUI programming. These tools are simple to use, yet powerful. Let's get to know them better.



Note: I used the Ubuntu 11.10 64-bit edition to test the examples presented in this article.

Quick and easy GUIs with EasyGUI

Tkinter is an event-driven tool-kit, and takes time to learn and program—but the EasyGUI wrapper hides all the rocket science. In this set of many general GUI dialogue boxes, each one has its own event loop, and is independent of the others. It's a more feature-rich, true graphics-mode counterpart of shell utilities like *dialog*, *whiptail*, etc. EasyGUI is a standalone single-file Python module, plus a demo application based on Tkinter. It has no dependency